MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) FIFTH SEMESTER

MAJOR ELECTIVE - FOOD PRESERVATION (MG5DSECND303)

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A

Multiple Choice Questions
Answer all questions.
Each question carries 1 mark

1. Which of the following is an emerging non-thermal method of food preservation? [U][1]

a) Canning
b) Pasteurization
c) High Pressure Processing (HPP)
d) Sun drying

2. Fermentation as a preservation technique works mainly by: [U][5]

a) Removing all moisture from food
b) Microbial action producing acids/alcohol that inhibit spoilage organisms
c) Freezing the food rapidly
d) Applying high pressure

3. The "hurdle technology" concept in food preservation refers to: [U][2]

a) Using a single preservation method at very high intensity
b) Combining multiple preservation factors at lower individual intensities to inhibit microbial growth
c) Physically hurdling food over obstacles during processing
d) A packaging technique only

4. Pasteurization differs from sterilization mainly in that pasteurization [U][3]

a) Destroys all microorganisms including spores

b) Uses lower temperatures to destroy pathogenic organisms, not all spores

c) Is used only for solid foods

d) Requires no heat at all

5. Dehydrofreezing is a combination of: [U][4]

a) Freezing followed by canning b) Partial dehydration followed by freezing

c) Fermentation followed by drying d) Pasteurization followed by chilling

6. Nitrite is particularly important in cured meat because it: [U][4]

a) Increases water activity b) Helps develop the characteristic cured colour

c) Increases microbial growth d) Removes all fat

7. Canning preserves food mainly through a combination of: [U][3]

a) Heat processing and hermetic sealing to prevent recontamination

b) Freezing and dehydration

c) High sugar concentration only

d) Smoking and salting

8. In foam mat drying, why is aerating the liquid important? [U][4]

a)It reduces the need for heat

b)It forms a stable foam, increasing surface area for faster drying

c)It makes the food edible without cooking

d)It removes all chemical contaminants

9. Which of the following foods is commonly preserved using solar drying? [U][4]

a)Milk powder b)Coffee powders

c)Fresh fruits and vegetables d)Protein supplements

10. Which of the following belongs to Class I preservatives? [U][5]

a) Sodium benzoate b)Sulphur dioxide

c) Salt, sugar, vinegar d) Sorbic acid

Part B
Short Answer Questions
Answer any 5 questions
Each question carries 2 marks

11. Explain the role of high sugar concentration in preserving jams and jellies. [An][4]
12. Define fermentation and mention two advantages of fermentation as a preservation method. [U][5]
13. Differentiate between chilling and freezing as methods of preservation. [A][3]
14. Distinguish between antioxidants and antimicrobial agents used as food preservatives. [U][5]
15. What is the purpose of blanching before canning or freezing vegetables? [U][3]
16. Define food preservation and state its importance. [U][1]
17. What is food irradiation? [U][2]

Part C
Short Essay Questions
Answer any 6 questions
Each question carries 5 marks

18. Describe Ohmic heating as a method of food processing. How is it different from conventional heating? [U][1]
19. Define food preservation and its principles [U][1]
20. Differentiate between pasteurization and sterilization with suitable examples of foods preserved by each. [An][3]
21. Explain freezing as a method of food preservation and its impact on food quality. [An][3]
22. Elaborate on Class II preservatives with examples. [U][5]
23. Describe foam mat drying and its application in the food industry. [A][4]
24. Discuss the working principle of spray drying and explain why it is preferred for liquid and heat-sensitive food products. [An][4]
25. Compare solar drying and mechanical drying in terms of process, cost, efficiency, and quality of the final product. [An][4]

Part D

Essay Questions

Answer any 2 questions 10 marks

26. Explain the emerging trends in food preservation. [U][1]

27. Define fermentation, explain its types with examples, and discuss its advantages as a natural method of food preservation. [U][5]

28. Explain the various methods of preservation by dehydration [An][4]

29. Analyse the different methods of low-temperature food preservation [An][3]